

CADABRA AI: Prompt-to-CAD Architecture Map

Detailed Workflow, LLM Reasoning, Native API Execution & System Results

STAGE 1 User Prompt Input (SOLIDWORKS Task Pane)

The engineer enters a natural-language engineering query directly into the CADABRA panel in SOLIDWORKS.

Example Prompt: "Add a 6mm counterbored hole pattern on a 40mm bolt circle and fillet outer edges with 2mm radius."



STAGE 2 Contextual CAD Parser & Local Metadata Extraction

Before reaching the AI, CADABRA inspects the active CAD environment locally:

- **Active Geometry State:** Extracts selected faces, planes, existing sketch features, coordinate axes, and active document units (mm / inches).
- **Standards Mapping:** Evaluates company standards (Hole Wizard profiles, materials, sheet metal gauges, custom properties).
- **Payload Packaging:** Securely packages prompt + structural CAD metadata over encrypted HTTPS/TLS.



STAGE 3 AI Copilot Engine & LLM Processing

The cloud engine processes the payload using high-reasoning AI models fine-tuned on parametric CAD logic:

- **AI / LLM Architecture:** Utilizes advanced frontier LLMs (e.g., Anthropic Claude / OpenAI GPT-4o fine-tunes).
- **Intent & Mathematical Parsing:** Converts plain English intent into geometric constraints, equations, and feature sequencing.
- **API Command Generation:** Synthesizes native C# / Visual Basic COM interop calls specifically tailored for the SOLIDWORKS API.



STAGE 4 Guarded Execution & Self-Healing Rebuild Loop

Commands execute directly inside SOLIDWORKS in real-time with continuous verification:

- **Live API Execution:** Creates reference planes, draws constrained sketches, executes extrusions/cuts, and applies mates.
- **Error Interception:** If SOLIDWORKS triggers a rebuild error or broken relation, CADABRA catches the failure stack.
- **Auto-Correction:** Adjusts parameters, rolls back state, and re-executes automatically until geometry resolves cleanly.



STAGE 5 Native Parametric Output Results

Generates native, fully manufacturing-ready CAD deliverables:

- **Parametric Feature Tree:** Full history of editable sketches, extrusions, revolves, patterns, and Hole Wizard features.
- **Production 2D Drawings:** Fully annotated 2D sheets with title blocks, GD&T, and automated Bill of Materials (BOM).

SUMMARY KEY TECHNICAL CHARACTERISTICS

LLM & AI Model Specs

- **Reasoning Models:** Anthropic Claude 3.5 Sonnet / OpenAI GPT-4o integrations.
- **Specialization:** Fine-tuned on SOLIDWORKS COM API schemas and mathematical constraint solving.
- **Security:** Enterprise zero-data retention pipelines.

Output vs. Mesh Generators

- **CADABRA:** Fully editable, parametric native ``.SLDPRT`` / ``.SLDASM`` feature trees.
- **Mesh Tools (Mesh AI):** Static uneditable polygonal mesh blobs (``.STL`` / ``.OBJ``).
- **Manufacturing:** Directly usable for CNC, tooling, and shop floor drawings.